

2024 Ph H2 Q11

Section: Particles and Waves

Topic: Spectra and Energy Levels

Summary of Question

The diagram shows four energy levels for hydrogen: E_0 , E_1 , E_2 , E_3 . Tasks: (a)(i) number of possible emission lines; (a)(ii) transition for the shortest wavelength; (b)(i) wavelength for transition $E_3 \rightarrow E_1$ and (ii) its colour; (c) explain why the red line is brighter.

(a)(i) Number of possible emission lines

With 4 levels, the number of distinct downward transitions is the number of unordered pairs: $nC_2 = 4C_2 = 6$.

Final answer: 6 lines

(a)(ii) Transition giving the shortest wavelength

Shortest wavelength $\leftrightarrow$ highest photon energy $\leftrightarrow$ largest energy gap.

From the diagram values ($|E_3 - E_0| >$ any other), the largest gap is from the highest to the ground level.

Final answer: $E_3 \rightarrow E_0$

(b)(i) Wavelength for $E_3 \rightarrow E_1$

Energy difference: $\Delta E = |E_3 - E_1| = |(-1.36 \times 10^{-19}) - (-5.45 \times 10^{-19})| = 4.09 \times 10^{-19} \text{ J}$.

Use $\lambda = hc/\Delta E$ with $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$ and $c = 3.00 \times 10^8 \text{ m s}^{-1}$.

$$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / 4.09 \times 10^{-19} \approx 4.863 \times 10^{-7} \text{ m}.$$

Final answer: $4.86 \times 10^{-7} \text{ m}$ ($\approx 486 \text{ nm}$)

(b)(ii) Colour of this photon

A wavelength of $\approx 486 \text{ nm}$ lies in the blue-green region of the visible spectrum.

Final answer: blue-green

(c) Why the red emission line is brighter

Brightness depends on how many photons of that wavelength are emitted per second.

The red line is brighter because more electrons undergo that particular transition per second (higher transition probability and/or greater population of the upper level), so more photons of that energy are produced.

Final answer: more transitions per second $\rightarrow$ more photons $\rightarrow$ brighter line

Revision Tips

For N energy levels, possible emission lines = $N(N-1)/2$.

Shortest $\lambda \leftrightarrow$ largest ΔE (usually highest level to ground).

Link energy and wavelength via $\Delta E = hc/\lambda$; always give units and sensible sig. figs.